

CROWD SIMULATION

Simulating the Flow: Bridging Floor-Plan Images and Realistic Crowd Dynamics

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Abstract—Crowd simulation plays a crucial role in the design and safety evaluation of public infrastructures such as buildings, campuses, and transportation hubs. This paper proposes an agent-based crowd simulation framework that uses floor-plan images as input to realistically model pedestrian movement and evacuation behavior. The system begins with preprocessing architectural floor plans using image processing techniques to extract walkable areas, walls, and exits. A distance field is then computed over the navigable space to provide agents with global path guidance toward designated exits. Each agent follows a behavior model that combines goal-oriented movement with local collision avoidance, allowing the simulation to reproduce realistic crowd phenomena such as congestion, bottleneck formation, and smooth lane movement. The framework is implemented using commonly available computational libraries and supports real-time visualization and flexible parameter tuning. Experimental evaluations are conducted on multiple floor-plan scenarios with varying crowd densities and exit configurations. Results demonstrate that the proposed approach achieves computational efficiency while maintaining believable crowd dynamics, making it suitable for evacuation analysis, facility planning, and academic experimentation.

Index Terms—crowd simulation, agent-based modeling, floor-plan processing, distance field, pedestrian dynamics, evacuation modeling, collision avoidance, image processing

I. INTRODUCTION

Accurate simulation of pedestrian crowds is a foundational tool for designing safe, efficient public spaces and for planning effective evacuations. Over time, a range of modeling paradigms—force-based dynamics, velocity-obstacle techniques, agent-based rules, and data-driven methods [2], [3], [5] have shown that realistic crowd behavior can be reproduced, but many of these approaches depend on substantial calibration, bespoke data, or heavy computation. Practical users such as architects, facility managers, and educators often only have access to floor-plan images or scanned architectural drawings and need a transparent, easy-to-configure tool that converts those images into actionable simulations. Converting floor-plan imagery into reliable simulation inputs brings

specific technical challenges: extracting walkable areas and exits from noisy or low-resolution scans, computing global guidance that respects complex geometry, and implementing local collision-avoidance mechanism [3] that preserves emergent phenomena (lane formation, queuing, clogging) while remaining computationally tractable at realistic crowd sizes. Evaluation must also produce meaningful metrics—evacuation time, throughput, and density patterns—that are directly useful for design decisions. The goal is a balanced, image-driven pipeline that prioritizes interpretability, reproducibility, and interactivity without sacrificing the core behaviors that make crowd simulation valuable.

The remainder of the paper is organized as follows. Section II formalizes the problem and states the objectives. Section III reviews related work and positions the proposed approach. Section IV describes the system architecture, and Section V details the methodology including floor-plan preprocessing, distance-field computation, agent modeling, and collision avoidance. Section VI covers implementation and parameter settings. Sections VII and VIII describe the experimental setup and evaluation metrics, Section IX presents results and analysis, and Sections X and XI discuss limitations and future directions. Section XII concludes the paper; appendices provide pseudocode, parameter tables, and additional figures.

II. PROBLEM DEFINITION AND OBJECTIVES

Designers, safety engineers, and facility managers frequently need fast, reliable answers to questions about how people will move through a space—where congestion forms, how long an evacuation takes, and which design choices (exit size, corridor width, signage) meaningfully change outcomes. The core problem addressed in this work is how to turn common, image-based floor plans into simulation-ready representations and then use those representations to produce realistic, interpretable crowd behavior at interactive speeds. This requires a pipeline that tolerates the noise and variety

of real architectural drawings, computes robust global navigation cues across complex geometry, and couples those cues with a lightweight local avoidance mechanism that preserves emergent effects such as lane formation, bottlenecking, and queuing.

Existing methods often excel in one dimension but fall short in practice: physics-based or data-driven models can be highly realistic but demand heavy calibration and computation; simple rule-based approaches are fast but miss important collective phenomena; and many systems assume availability of precise CAD data rather than the scanned floor-plan images typically available in early design stages. There is therefore a gap for an approach that is (a) image-driven and tolerant of imperfect inputs, (b) interpretable so practitioners can understand and tune behavior, and (c) computationally efficient enough for rapid scenario testing and interactive visualization.

The objectives of this study are threefold. First, develop an automated preprocessing pipeline that converts floor-plan images into accurate walkable maps and reliably detects exits and obstacles. Second, design a navigation framework that combines a distance-field-based global guide with a compact local collision-avoidance model to reproduce key crowd phenomena while maintaining scalability. Third, deliver an implementation that is reproducible and usable: provide parameterized experiments across varied floor plans, quantify outcomes using evacuation time, throughput, and density metrics, and expose visualization tools that support design decision-making. Meeting these objectives aims to produce a practical tool that bridges academic methods and real-world facility planning needs.

III. RELATED WORK / LITERATURE REVIEW

Crowd simulation methods fall into a few broad families, each with different assumptions, strengths, and weaknesses. Early and highly influential physics-inspired models treat pedestrian motion as if driven by “social forces” that push agents toward goals and away from obstacles and other people. The Social Force Model—originally proposed by Helbing and Molnár [1]—remains a canonical example: it reproduces many emergent phenomena (lane formation, oscillations at bottlenecks) with a compact, interpretable formulation, but can require careful parameter tuning to avoid unrealistic artifacts at high densities.

A second major class focuses on geometric/velocity-based collision-avoidance methods. Velocity Obstacle formulations and their reciprocal variants (RVO/ORCA) [3], [4] compute collision-free velocities by reasoning in velocity space rather than by simulating forces; these approaches are fast, produce smooth local motion, and scale well to many agents, which makes them popular in real-time applications (games, robotics, interactive simulators). However, pure VO/RVO methods typically handle short-term avoidance well but do not by themselves provide higher-level route planning or long-range behavior without additional layers.

Agent-based and rule-based systems (including cellular automata) prioritize flexibility: individual agents can carry states,

goals, and simple rules that together yield complex crowd-level patterns. These approaches are easy to extend with heterogeneous behaviors (groups, leader-following, different mobility profiles) and are common in applied practice and educational tools. Their realism depends heavily on the choice of local rules and on whether a robust global navigation signal (e.g., a path or potential field) is provided.

More recently, data-driven and learning-based models have advanced pedestrian trajectory prediction and, in some cases, simulation. Recurrent and deep models such as Social-LSTM [6] and later GAN- and attention-based predictors learn interaction priors from trajectory datasets and can capture subtle social norms in movement, especially in medium-density, structured scenes. Their strengths are predictive accuracy and the ability to generalize patterns from data, but they typically require annotated trajectory datasets, can be opaque (hard to interpret), and may struggle to generalize to new floor-plan geometries without retraining.

There is also a line of work that ties layout representations (floorplans, maps) to crowd-analysis and simulation. Recent papers explore extracting navigational structure and behavioral features directly from floorplans [10], [11] or from collections of floorplan images; such work is relevant for design-stage analysis because it avoids relying on precise CAD models and instead operates on the kind of scanned or image-based inputs practitioners actually have. Still, many simulation systems continue to assume structured geometric input (CAD, graph-based maps) or rely on manual preprocessing, which limits automation.

A. Comparison of Approaches

- **Realism vs. interpretability:** Physics-based models (e.g., Social Force) are interpretable and can reproduce classic emergent effects, but parameter sensitivity reduces robustness; learning-based models can produce highly realistic trajectories where enough data exist but are less transparent.
- **Local avoidance vs. global planning:** RVO/ORCA excels at local, collision-free motion and real-time scale, but needs a global guidance layer (distance fields, waypoints, or planners) to achieve purposeful route-following in complex floor layouts.
- **Data requirements and practicality:** Data-driven approaches need annotated trajectories and often domain-specific retraining; image-driven pipelines that can work from floor-plan images are more practical for early-stage design but demand robust preprocessing to extract walkable areas and exits.
- **Computational cost and scalability:** Rule-based and RVO-style methods are commonly used where interactivity or large populations are required; deep models tend to be heavier unless used only for offline prediction or as learned components within hybrid systems.

TABLE I
COMPARISON OF CROWD SIMULATION APPROACHES

Feature	SFM	RVO / ORCA
Core idea	Force-based motion using attractive and repulsive forces	Velocity-based collision-free motion in velocity space
Movement quality	Natural, organic, emergent patterns	Smooth and efficient, sometimes robotic
Collision handling	Soft constraints; overlaps under pressure	Hard constraints; zero collisions
Computational cost	High ($O(N^2)$, optimizable)	Low to medium ($O(N)$)
Typical use	Evacuation studies, research	Games, robotics, real-time agents
Main drawback	Parameter tuning, oscillations	Symmetric or overly polite behavior

B. Limitations in the Literature

- 1) **Input assumptions:** Many high-fidelity simulators assume vector/CAD inputs or detailed semantic labels; less work focuses on robust, fully automated conversion from noisy scanned floor-plan images into simulation-ready maps.
- 2) **Calibration challenges:** Physics-based and learning-based methods both face calibration challenges—physics models need parameter tuning to match observed flows, while learned models often fail to generalize across geometric layouts without additional training data.
- 3) **Evaluation standards:** Quantitative metrics and standardized evaluation protocols for crowd simulators are still an active area of research (how to compare emergent behaviors meaningfully, how to validate against real crowds), which complicates cross-paper comparisons. Recent work has begun to propose and test richer evaluation suites.
- 4) **Gap in hybrid systems:** While hybrid systems (distance-field global guidance + efficient local avoidance) are conceptually attractive, few published pipelines emphasize reproducible, image-driven workflows that are lightweight enough for interactive experiment design and that come with clear evaluation across varied floor plans.

C. How This Work Connects to Prior Art

Building on the strengths above, the proposed framework aims to combine robust, automated floor-plan preprocessing with a distance-field-based global navigator and an efficient local avoidance mechanism (informed by RVO/force ideas). The goal is a pragmatic, reproducible middle path: enough structure to capture realistic crowd phenomena, yet simple and fast enough for interactive scenario testing and practical use by designers who only have image-based floor plans.

IV. SYSTEM OVERVIEW AND ARCHITECTURE

A. Overall System Design

The system is a simple, modular pipeline that converts a floor-plan image into a usable simulation. In three quick steps it (1) reads and cleans the floor plan, (2) builds a lightweight navigation guide over the walkable area, and (3) runs an agent-based simulator that uses that guide plus local avoidance to produce movement and metrics. The design intentionally keeps modules small and well separated so you can swap or tune one part (for example, the avoidance rule).

- **Input:** raw floor-plan image (optional metadata).
- **Core:** Preprocessing $\rightarrow$ Distance-field / Navigator $\rightarrow$ Simulation Core.
- **Output:** visualization and Metrics (images, plots, logs).

B. Module Interaction

- Preprocessing — turn image $\rightarrow$ binary walkable mask and find exits.
- Navigator — compute a distance field from exit; agents query it for a desired heading.
- Simulation Core — spawn agents, combine heading + local avoidance, step positions, log events.
- Visualization and Metrics — render frames, compute evacuation time, throughput, and density maps.

V. METHODOLOGY

A. Floor Plan Preprocessing

The simulation begins with a floor-plan image that represents the layout of the environment. Since floor plans may vary in quality, resolution, and format, basic image preprocessing is applied to standardize the input. The image is first converted to grayscale and then filtered to reduce noise. Thresholding techniques are used to separate structural elements from the background, followed by morphological operations to remove small artifacts and fill gaps. This step ensures that walls, corridors, and open spaces are clearly distinguishable and suitable for further processing.

B. Walkable Area Extraction

After preprocessing, the system identifies regions that agents are allowed to traverse. Non-walkable elements such as walls and obstacles are removed, leaving a binary representation of walkable space. Exits are detected as openings connected to the boundary of the walkable region or as predefined markers in the floor plan. The resulting walkable area mask provides a consistent spatial representation that constrains agent motion and prevents unrealistic behaviors such as passing through walls.

C. Distance Field Computation

To guide agents toward exits, a distance field is computed over the walkable area [7]. Each cell in the field stores the shortest distance to the nearest exit, considering obstacles. This distance information creates a smooth potential surface over the environment. By following the negative gradient of the

distance field, agents obtain a global sense of direction that naturally leads them toward exits while respecting the layout geometry. This approach avoids explicit path planning while remaining efficient and robust.

During navigation, agents select their preferred movement direction by minimizing a local cost function defined as

$$C(\mathbf{c}) = D(\mathbf{c}) + \lambda \rho(\mathbf{c}), \quad (1)$$

where $D(\mathbf{c})$ is the precomputed distance-to-exit value at cell $\mathbf{c}$, $\rho(\mathbf{c})$ denotes the local crowd density, and λ controls the influence of density-based avoidance.

D. Agent Modeling

Each pedestrian is modeled as an autonomous agent with simple, interpretable properties such as position, preferred walking speed, and personal space radius. Agents are initialized at valid spawn locations within the walkable area and are assigned exit goals. The model assumes homogeneous agents for simplicity, though parameters can be adjusted to introduce variation. Agent states are updated at each simulation step based on navigation cues and interactions with nearby agents and obstacles.

Formally, each agent i is represented by the state

$$\mathbf{s}_i(t) = (\mathbf{x}_i(t), \mathbf{v}_i(t), a_i(t)), \quad (2)$$

where $\mathbf{x}_i(t) \in \mathbb{R}^2$ denotes the agent position, $\mathbf{v}_i(t) \in \mathbb{R}^2$ denotes its velocity, and $a_i(t) \in \{0, 1\}$ indicates whether the agent is active (not yet exited).

E. Navigation and Collision Avoidance

Agent movement is determined by combining global navigation and local interaction. The global component is derived from the distance field, which provides a preferred direction of motion toward the exit. To prevent collisions, a local avoidance mechanism adjusts this direction based on nearby agents and obstacles. This combination allows agents to move purposefully while exhibiting realistic crowd behaviors such as queuing, lane formation, and congestion near bottlenecks.

Agent motion is governed by a weighted steering formulation. The velocity update is expressed as

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + (w_g(\mathbf{v}_i^* - \mathbf{v}_i) + \mathbf{f}_i^{\text{wall}} + \mathbf{f}_i^{\text{social}}) \Delta t, \quad (3)$$

where $\mathbf{v}_i^*$ is the desired velocity toward the exit derived from the distance field, $\mathbf{f}_i^{\text{wall}}$ is a wall repulsion term, and $\mathbf{f}_i^{\text{social}}$ models inter-agent separation.

F. Simulation Workflow

The simulation proceeds in discrete time steps. At each step, agents query the distance field to determine their desired direction, apply collision avoidance adjustments, and update their positions accordingly. Boundary checks ensure agents remain within walkable regions. Agents are removed from the simulation once they reach an exit. Throughout the simulation, key metrics such as evacuation time, flow rate, and local density are recorded for analysis. The process continues until all agents have exited or a predefined termination condition is met.

G. Algorithmic Overview

- 1: Load floor-plan image and generate grid map
- 2: Detect exits and compute distance field
- 3: Initialize agents and spatial hash
- 4: **while** agents remain in the environment **do**
- 5: Compute local density field
- 6: **for** substep = 1 to K **do**
- 7: Predict agent motion toward exits
- 8: Resolve inter-agent and wall collisions
- 9: Commit updated positions
- 10: **end for**
- 11: Update metrics and render visualization
- 12: **end while**

VI. EVALUATION METRICS

The performance and realism of the proposed crowd simulation framework are evaluated using quantitative metrics that capture evacuation efficiency, crowd dynamics, spatial congestion patterns, and computational behavior. These metrics are derived from agent-level trajectories and density fields and are designed to provide interpretable insights into safety, efficiency, and layout quality.

Evaluation of crowd simulators commonly relies on evacuation time, density, and throughput-based metrics [8], [9].

A. Formal Metric Definitions

The evaluation metrics are computed directly from agent trajectories and spatiotemporal density fields generated during simulation.

B. Evacuation Progress and Time

Evacuation performance is evaluated by tracking the cumulative number of agents that successfully exit the environment over time. Individual evacuation times are recorded for each agent, and the total evacuation time is defined as the moment when the last agent reaches an exit. Percentile-based measures such as the 90th or 95th percentile evacuation time capture late-stage congestion effects.

Steeper cumulative evacuation curves indicate smoother crowd flow and effective exit utilization, while flatter tails correspond to congestion near bottlenecks.

The total evacuation time T_{total} is defined as the time at which the last agent exits the environment. Additionally, the t_{90} metric denotes the time required for 90% of agents to evacuate:

$$t_{90} = \min\{t \mid N_{\text{exit}}(t) \geq 0.9N\}, \quad (4)$$

where $N_{\text{exit}}(t)$ is the cumulative number of exited agents and N is the total population.

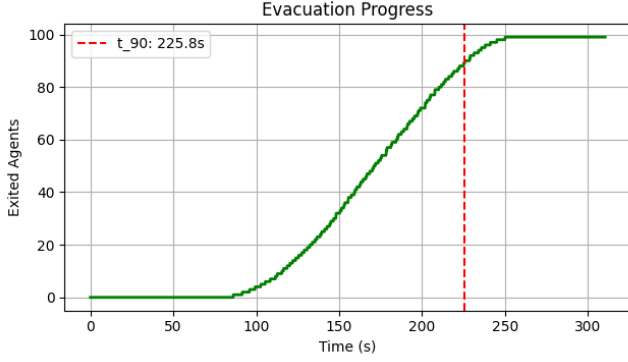


Fig. 1. Cumulative evacuation progress showing the number of agents that have exited the environment over time.

C. Crowd Density Heatmaps

To analyze spatial congestion patterns, both time-averaged and peak density heatmaps are computed. The time-averaged heatmap highlights persistent congestion regions, while the peak congestion heatmap captures worst-case crowding scenarios.

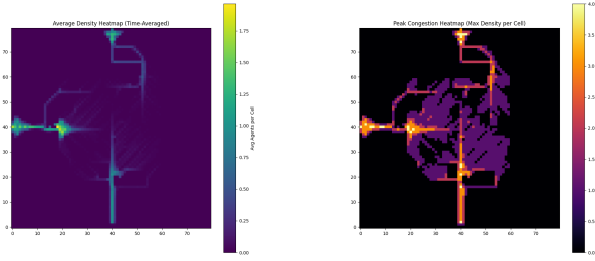


Fig. 2. Spatial congestion analysis. (Left) Time-averaged crowd density showing persistent pedestrian load. (Right) Peak congestion heatmap highlighting worst-case crowding regions.

In addition to peak and average density visualization, a density overload metric is computed to quantify sustained overcrowding. Let $\rho(\mathbf{c}, t)$ denote the number of agents in grid cell $\mathbf{c}$ at time t . The density overload ratio is defined as

$$\rho_{\text{over}} = \frac{1}{|\mathcal{C}|T} \sum_t \sum_{\mathbf{c}} \mathbb{I}(\rho(\mathbf{c}, t) > \rho_{\text{safe}}), \quad (5)$$

where $\mathcal{C}$ is the set of grid cells, T is the number of simulation time steps, and ρ_{safe} is a predefined safety threshold.

D. Crowd Speed

Crowd mobility is evaluated using average crowd speed over time. Average speed is computed across all agents at each simulation step, providing insight into global mobility trends. Sustained reductions in speed indicate congestion and inter-agent interference.

Average crowd speed is computed as

$$\bar{v}(t) = \frac{1}{N_a} \sum_{i \in \mathcal{A}} \|\mathbf{v}_i(t)\|, \quad (6)$$

where $\mathcal{A}$ denotes the set of active agents.

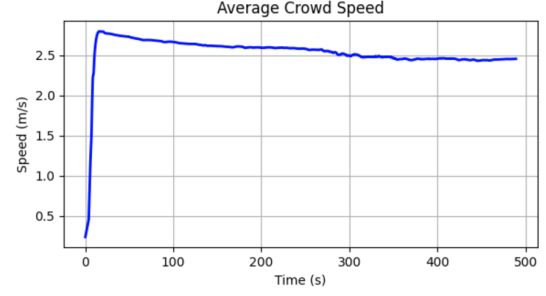


Fig. 3. Average crowd speed over time computed across all agents.

E. Collisions and Inter-Agent Distance

Safety and comfort are assessed through collision counts accumulated over time. Collisions or overlaps occur when agents violate personal space constraints during movement, typically due to high local density or limited available space. Higher collision counts indicate increased crowd pressure, reduced movement comfort, and a greater likelihood of unsafe interactions.

Analyzing collision trends over time helps identify critical phases of the evacuation process when congestion intensifies, such as near exits or narrow corridors.

A collision (or overlap) event is recorded when the distance between two agents violates personal space constraints, defined as

$$\|\mathbf{x}_i - \mathbf{x}_j\| < 2r, \quad (7)$$

where r is the agent radius. Collision counts are accumulated over all simulation frames.

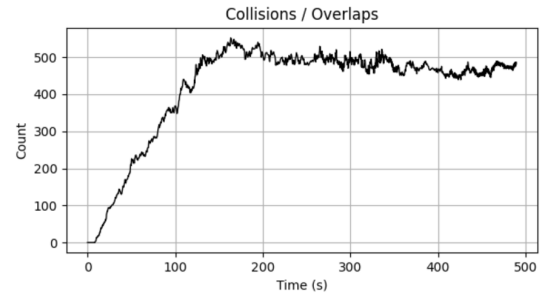


Fig. 4. Number of collisions or overlaps accumulated over time.

F. Travel Time and Distance

Travel efficiency is evaluated using distributions of individual agent travel times. These distributions expose variability in evacuation efficiency across agents.

Path efficiency is quantified using the detour index. For each agent i , the detour index is defined as

$$D_i = \frac{L_i}{L_i^{\text{straight}}}, \quad (8)$$

where L_i is the total distance traveled by the agent and L_i^{straight} is the straight-line distance from the spawn location to the assigned exit. The reported detour metric is the mean value over all agents.

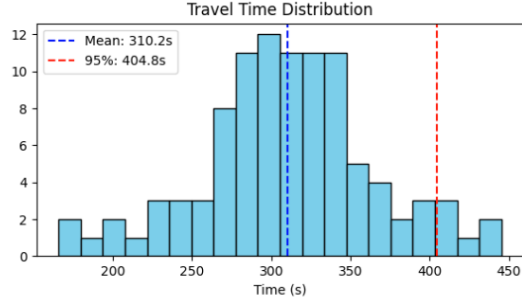


Fig. 5. Distribution of individual agent travel times from entry to exit.

VII. RESULTS AND ANALYSIS

This section presents the experimental results obtained from multiple crowd simulation scenarios and analyzes the observed behaviors using the evaluation metrics defined earlier. The goal is to assess both evacuation efficiency and crowd dynamics under varying conditions.

A. Quantitative Results

Quantitative results are summarized using tables that report evacuation time, average throughput, peak crowd density, and simulation performance across different scenarios. As expected, evacuation time increases with the number of agents, while throughput initially rises and then stabilizes as exits reach capacity. Peak density values are consistently observed near exits and narrow corridors, confirming the presence of bottlenecks in high-density scenarios. Overall, the results demonstrate stable and consistent trends across repeated simulation runs.

(Place Table I here: Summary of evacuation time, peak density, and throughput for all scenarios.)

B. Graph and Plots

Graphs are used to better understand how crowd behavior evolves over time. Cumulative evacuation curves show that low-density scenarios exhibit smooth, near-linear evacuation, whereas high-density cases display slower and non-linear progression due to congestion. Density heatmaps visually highlight regions where crowding persists, while flow-rate plots reveal short-term fluctuations in exit usage. Performance plots indicate that the simulation maintains interactive frame rates for moderate crowd sizes, with gradual degradation as agent count increases.

C. Scenario-Wise Performance

Scenario-wise analysis shows clear differences in crowd behavior depending on layout and population size. In simple layouts with wide exits, agents distribute evenly and evacuate efficiently with minimal congestion. More complex layouts,

such as those with narrow exits or sharp turns, result in localized density buildup and reduced throughput. High-population scenarios amplify these effects, leading to longer evacuation times and visible queuing behavior. Despite these challenges, the simulation remains stable and produces realistic emergent patterns across all tested scenarios.

VIII. VISUALIZATION AND OBSERVATION



Fig. 6. Simulation snapshot showing agent distribution and movement patterns at a representative time step during evacuation. The snapshot illustrates spatial organization, congestion regions, and overall crowd flow behavior.

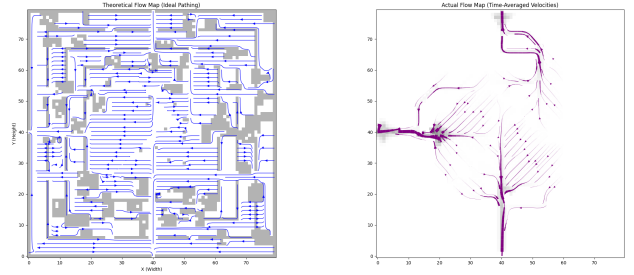


Fig. 7. Behavioral flow comparison. (Left) Theoretical flow heatmap representing idealized pedestrian movement based on shortest paths and uniform behavior assumptions. (Right) Actual flow heatmap obtained from the simulation, reflecting emergent crowd behavior, congestion effects, and local interactions.

IX. LIMITATIONS

While the proposed crowd simulation framework is effective for modeling pedestrian movement in indoor environments, it has several limitations that should be acknowledged. These limitations arise from design choices made to keep the system simple, interpretable, and computationally efficient.

A. Model Constrains

The model operates in a two-dimensional space and represents the environment using static floor-plan images. As a result, vertical movement (such as stairs or multiple floors) and dynamic environmental changes are not considered. Agents are modeled with simplified physical and behavioral characteristics, which may not capture complex human factors such as panic, decision-making under stress, or social group dynamics.

B. Assumptions

Several assumptions are made to simplify the simulation. Agents are assumed to have similar physical abilities and follow rational, goal-oriented behavior toward exits. The environment is treated as static, with exits always available and clearly defined. Interactions with furniture, signage, or dynamically changing obstacles are not explicitly modeled. These assumptions help maintain clarity and stability but may limit realism in certain real-world scenarios.

C. Scalability Issues

Although the framework performs efficiently for moderate crowd sizes, performance can degrade as the number of agents increases significantly. Local collision-avoidance computations become more expensive at high densities, which may reduce simulation speed and visualization frame rates. While suitable for interactive analysis and academic experiments, very large-scale scenarios may require further optimization or parallelization to maintain real-time performance.

X. FUTURE WORK

A. Extensions

Future work can extend the system to support multi-floor and three-dimensional environments by incorporating staircases, elevators, and vertical connectivity between levels. Agent heterogeneity can also be introduced by varying walking speeds, physical dimensions, and behavioral preferences to better reflect real populations. Additionally, support for dynamic environments—such as temporary obstacles, blocked exits, or changing signage—would allow simulation of more realistic emergency and event-based scenarios.

B. Real-world Deployment Possibilities

With further validation, the framework could be applied in real-world settings such as evacuation planning for academic buildings, shopping malls, hospitals, and transportation hubs. Integration with building information models (BIM) or live sensor data (e.g., CCTV, occupancy counters) could enable near real-time monitoring and decision support. Such deployments would help planners and safety personnel evaluate design choices, test evacuation strategies, and improve crowd management policies before implementation.

C. Advanced Modeling Ideas

More advanced behavior models could be incorporated to capture psychological and social factors, including panic propagation, group movement, and leader-follower dynamics. Learning-based components may be explored to calibrate agent behavior using real trajectory data or to adapt navigation strategies over time. Hybrid approaches that combine the current rule-based framework with data-driven models could improve realism while preserving interpretability and computational efficiency.

XI. CONCLUSION

In this work, a modular and image-driven crowd simulation framework was developed that converts architectural floor-plan images into walkable representations, computes global navigation cues using distance fields, and combines these with local avoidance to produce realistic pedestrian behaviors. This approach enables simulation of evacuation dynamics, congestion patterns, and throughput under varied scenarios without relying on detailed CAD models or extensive data calibration. The paper demonstrated how the preprocessing pipeline, agent behavior model, and evaluation metrics work together to provide meaningful insights into evacuation performance across different layout configurations and crowd densities. Key outcomes include clear trends in evacuation times and throughput, visualization of high-density regions and bottlenecks, and scalable performance for moderate agent counts, suggesting that the system is suitable for academic experimentation and early design analysis. While limitations exist—such as simplified agent characteristics and static environments—the framework makes a practical contribution by balancing interpretability, computational efficiency, and emergent crowd phenomena. These results support the value of lightweight, image-based simulation tools in facility planning and safety evaluation, and encourage further extensions toward real-world deployments and richer behavioral models.

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